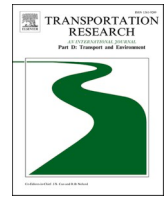




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Energy consumption models for delivery drones: A comparison and assessment

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ABSTRACT

Energy consumption is a critical constraint for drone delivery operations to achieve their full potential of providing fast delivery, reducing cost, and cutting emissions. This paper provides a uniform framework to facilitate understanding different drone energy consumption models and the inter-relationships between key factors and performance measures to facilitate decision making for drone delivery operations. We review, classify and assess drone energy consumption models. We then document the very wide variations in the modeled energy consumption rates resulting from differences in: (1) the scopes and features of the models; (2) the specific designs of the drones; and (3) the details of their assumed operations and uses. The results show that great care must be taken in adopting a particular drone energy consumption model and that more research is needed, especially empirical research, to ensure the selected model accurately reflects delivery drone designs and uses.

1. Introduction

Unmanned Aerial Vehicles (UAVs) or drones have been proposed as a method for delivering goods to consumers with benefits of lower cost, increased speed, and reduced greenhouse gas emissions. For drone delivery, the drone energy requirements determine the key performance metrics of range (or endurance), cost and emissions. An accurate estimation of drone energy consumption ensures feasible as well as efficient operating decisions for drone delivery systems, as well as for all drone applications including surveillances, wireless communications, inspections, agricultural operations, humanitarian assistance missions and others. However, many optimization models that design drone or truck-drone routes or drone delivery systems incorporate energy consumption only indirectly as a fixed limit on drone endurance (flight time limit) or range (flight distance limit) (e.g., Murray and Chu, 2015; Chiang et al., 2019; Kithacharoenchai et al., 2020). Other drone delivery research incorporates energy directly with an energy consumption model based on the fundamental physical forces involved in flight or on field measurements (e.g., Kirchstein, 2020; Murray and Raj, 2020; Poikonen and Golden, 2020; Stolaroff et al., 2018; Figliozzi, 2017; Dorling et al., 2017). Some of these drone energy consumption models are quite simple with only a few parameters, while others are very complex comprised of multiple interdependent components that provide detailed representations of the forces of flight and drone design.

However, these various drone energy models can produce widely divergent results in terms of the energy consumed for essentially the same drone delivery operations, leading to wide differences in modeled drone ranges and emissions. These differences create the

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Drone design	Environment	Drone dynamics	Delivery Operations
• Drone weight • Number of rotors • Size of rotors • Size of drone body • Battery weight • Battery energy capacity • Size of battery • Power transfer efficiency • Maximum speed and payload • Lift-to-drag ratio • Delivery mechanism • Avionics	• Air density • Gravity • Wind conditions • Weather (rain, snow, etc.) • Ambient temperature • Regulations	• Airspeed • Motion (takeoff / landing, hover, cruise) • Acceleration and Deceleration • Angle of attack • Flight angle • Flight altitude • Riding on another vehicle	• Payload weight • Size of payload • Empty return • Fleet size and mix • Single-/multi-stop drone trip • Delivery mode (landing, tether, parachute) • Area of service region

Fig. 1. Factors affecting drone energy consumption.

need to carefully delineate why such differences exist when modeling the same phenomena (i.e. drone flights or drone delivery) and to assess the different approaches to modeling drone energy consumption. While we limit our consideration to models of battery-powered aerial drones, such as those proposed for home delivery (Lee, 2019; Josephs, 2019) and related activities (e.g., medical deliveries over short ranges (Cohen, 2019; Drones in HealthCare, 2020)), our analyses and insights are also applicable and important to other applications of drones with limited energy capacity.

The wide variation of energy consumption in published drone delivery research is a result of different scopes and features of the models, different designs of the drones being modeled, and different assumed operations. Thus, current research has not reached consensus on standards for drone energy consumption, nor on how to model delivery drone energy consumption – and therefore existing models may not reflect drone delivery operations well. As drone technology is evolving rapidly, our work strives to help improve understanding of drone energy consumption and to develop common standards. For conventional ground vehicles such as trucks and cars, we have good fuel consumption standards thanks to decades of rigorous research from government agencies, universities and manufacturers (Barth et al., 2005). Given that the operations of drones are more sensitive to the energy capacity than those of conventional vehicles (Cheng et al., 2020), it is critical to develop good understanding and estimation of drone energy consumption.

The key contributions of this paper are to: (1) review and classify key drone energy consumption models using a unified notation that allows a collective comparison of the energy consumption rates for the various drone types, payloads, speeds, etc.; (2) examine how the distinguishing features of different drone energy consumption models contribute to differences in the calculated energy consumption; (3) evaluate published results on energy consumption for small delivery drones from both the models and field tests; and (4) compare the drone energy consumption models in a common setting for two prototypical drones to document how energy use and range vary differentially as a function of speed and payload. We believe the results provide useful insights for modeling energy consumption for current drones, as well as for future (not yet existing) drones.

The rest of the paper is organized as follows. Section 2 provides background and classification of drone energy consumption models in the literature. Section 3 discusses key drone energy consumption models using a unified notation. Section 4 documents the differences in the energy consumption rates as reported in the literature as well as in a common setting. Section 5 discusses insights and implications of the analyses, and Section 6 provides concluding remarks.

2. Background and features of drone energy consumption models

2.1. Background

Key factors affecting drone energy consumption can be classified into four categories: drone design, environment, drone dynamics, and delivery operations. Major factors in these four categories are provided in Fig. 1 (adapted from Demir et al. (2014) for road transport). Drone design factors include the weight and size of the drone body, the number and size of rotors, the weight, size and energy capacity of the battery, power transfer efficiency, maximum speed and payload, lift-to-drag ratio, delivery mechanism and avionics. Environmental factors include air density, gravitational force, wind conditions, weather (snow, rain, etc.), ambient temperature and operating regulations. Drone dynamics factors include drone travel speed, drone motion (i.e., takeoff/landing, hover,

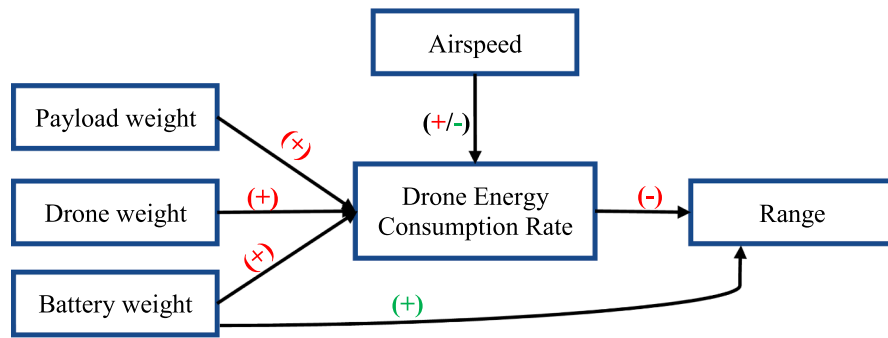


Fig. 2. Interrelated aspects of the energy consumption rate for drone delivery.

horizontal flight), acceleration/deceleration, angle of attack and flight altitude. We also include the possibility for drones to be carried on other vehicles such as trucks or public transit, for a portion of their trip. Delivery operations factors include weight and size of the payload, “empty returns” (i.e. the return trip after delivery is without the payload, which implies a successful delivery), fleet size and mix, the number of deliveries per trip, the delivery mode, and the area of the service region. Some of these factors are determined or limited by the drone design (e.g., maximum payload, projected area of the drone, etc.), others are operational factors that can vary for a given drone design (e.g., payload, speed, etc.), and still others are external factors (e.g., weather). Further, many of these factors are interdependent and dynamic during a drone delivery trip.

Uncertainty in estimating drone energy consumption can result from all of these factors, especially drone design, drone dynamics, and delivery operations. Note that if drones are carried on other vehicles (e.g., trucks or public transit) they expend little or no energy, and that different payload delivery modes (landing, lowering via tethers, parachuting) require different levels of energy use. When drones are paired with other vehicles, or multiple drones are operating closely together (as for take-off and landing from the same site), considerable coordination and synchronization may be required, which might lead to substantial hovering by the drone(s).

A drone uses energy to fly by generating thrust and lift forces to overcome the weight and drag forces. Fig. 2 highlights five key interrelated aspects of drone energy consumption: payload weight, battery weight, drone (airframe) weight, airspeed, and range. The airspeed, payload, drone and battery weight are important determinants of the drone energy consumption rate. The energy consumption rate in turn, along with the battery weight (and type), determines the range. Note that the payload, drone and battery weight determine the total weight of the drone at takeoff, and involve both design decisions and operating decisions. Increasing the weight of any component increases the energy consumption rate, *ceteris paribus*, as indicated by the “(+)” on the three arrows from the weights to the energy consumption rate. Airspeed is an operating decision, and for drones (as well as airplanes and helicopters) the power consumed is approximately a convex function of airspeed (due to the competing forces of induced drag, parasite drag, and profile drag – see for example Rotaru and Todorov (2017)). Thus, there is a “(+/-)” on the arrow out of Airspeed. Range is determined by both the drone design and operating decisions. Note that an increase in battery weight will increase the drone energy consumption rate, *ceteris paribus*, which decreases the range; but the larger battery will also increase the available energy capacity, which increases the range (see Stolaroff et al. (2018) for a discussion of how battery affects drone energy use and range).

2.2. Relevant literature and key drone energy models

Since the seminal work of Murray and Chu (2015), many researchers have considered designing and optimizing drone delivery, with drones launched from a depot and/or other vehicles (usually trucks). This research includes drone routing and scheduling (e.g., Dorling et al., 2017; Agatz et al., 2018; Schermer et al., 2019; Liu, 2019; Murray and Raj, 2020; Kithacharoenchai et al., 2020), facility location problems including charging stations (Chauhan et al., 2019; Shavarani et al., 2018; Hong et al., 2018; Ferrandez et al., 2016), and fleet sizing (Troudi et al., 2018). For recent surveys of drone modeling which especially focus on combined truck-drone operations where drones can be carried by trucks to extend their useful range (and in delivery settings where the trucks serve as resupply points for drone deliveries), see Otto et al. (2018) and Chung et al. (2020). Other work that allows drones to extend their useful delivery range by riding on public transit includes Huang et al., 2020a, 2020b; Choudhury et al., 2019.

Drone travel models generally impose time and/or distance limit constraints to reflect the drone’s limited battery capacity. Much of the research assumes a constant rate of energy consumption per unit time or unit distance, so that drone energy consumption is modeled as a linear function of time or distance traveled (e.g., Ferrandez et al., 2016; Ha et al., 2018; Huang et al., 2020a; Moore, 2019). However, there is considerable variance in the assumed consumption values; for example, Ferrandez et al. (2016) use a value of 46.1 J/m (based on a 2013 Amazon delivery drone carrying 5 lb packages at 70 km/hr), while Moore (2019) uses 223.7 J/m (based on field tests with a DJI Matrice 600 Pro by the U.S. Department of Energy 2019, 2020). Rather than using a single fixed energy consumption rate, Goodchild and Toy (2018) consider energy consumption rates ranging from 22 to 223 J/m (10–100 Wh/mile) to assess the sensitivity of the research findings to the energy efficiency of drones. Their results show how the emissions benefits of drones relative to trucks strongly depend on the drone energy consumption rate; however, they do not suggest a particular energy consumption rate to use.

Some optimization models have incorporated drone energy consumption models explicitly, with a key differentiation being the

assumption regarding thrust for steady horizontal flight. The model may assume that (i) the thrust force (T) equals the drag force (D), and the weight force (W) equals the lift force (L) (therefore, $T = W/r$, where $r = L/D$ is the lift-to-drag ratio), (ii) the thrust force equals the weight force, as for a hovering helicopter, or (iii) the thrust force equals the sum of the weight, the drag and the lift forces. These different assumptions reflect different perspectives on the drone operations, e.g., whether they operate more like fixed-wing aircraft or helicopters. These three approaches give rise to three continuing streams of literature for drone energy modeling. Key references are described briefly in this section, with details in [Section 3](#).

[D'Andrea \(2014\)](#) provides a seminal contribution in modeling drone energy consumption by translating the fundamental flight principles of manned aircraft to a model for the much smaller scale of unmanned aerial drones. This article presents a model using an integrated approach that combines aerodynamic and drone design aspects into a single critical parameter: the lift-to-drag ratio r . The energy model also includes a fixed component for avionics power. [Troudi et al. \(2018\)](#) uses the same model to analyze drone fleet size, though they ignore the avionics power. [Figliozzi \(2017\)](#) adopts the same modeling approach to derive drone emissions based on a continuous approximation travel distance model. The D'Andrea integrated model is also used in a series of reports from the RAND Corporation that explore energy consumption for city-scale drone delivery systems ([Lohn, 2017; Xu, 2017; Gulden, 2017](#)). [Lohn \(2017\)](#) uses this model to analyze truck and drone delivery in cities of various sizes, and [Gulden \(2017\)](#) provides a GIS-based analysis of shifting truck deliveries to drones in Minneapolis. [Xu \(2017\)](#) discusses vehicle design aspects related to drone energy consumption and suggests that fixed-wing VTOL (vertical takeoff and landing) or hybrid multicopter configurations that combine VTOL capabilities with lifting surfaces (wing-like structures) are a better fit for many drone delivery operations. Several such drones have been tested, including the latest delivery drone designs of Amazon, Google, and others ([Lee, 2019](#);

[Heath, 2018; Adams, 2016; Swoop Aero, 2019](#)).

A second approach for modeling drone energy consumption is a component model based on helicopter operations, with the assumption that the power consumed during level flight, takeoff, or landing is approximately equivalent to the power consumed while hovering. This model ignores the impact of drone speed on energy consumption. [Dorling et al. \(2017\)](#) provides an equation for the power consumed by a multirotor helicopter in hover as a function of the battery and payload weight. These authors also report field experiments and develop regression parameters with small payloads (see [Section 3.3](#)). [Jeong et al. \(2019\)](#) adapts Dorling's hovering model for a MikroKopter MK8-3500 drone, and also describe a regression model based on payload mass. They state the "proposed energy consumption model provides realistic values that are analogous to the experiment result"; however, they do not provide any parameter values for the model.

A more detailed component modeling approach is comprised of component models based on the fundamental forces of flight, including the weight force of the aircraft (due to gravity) and drag forces. The two main drag forces acting on aircraft are parasite drag from the aircraft moving through the atmosphere and induced drag from redirecting the airflow to create the lift that keeps the aircraft aloft. Component models for drone energy consumption include separate models for the individual forces and the different components of a drone trip (takeoff, landing, cruising, hovering), and are often quite detailed to capture particular features of the drone design. [Stolaroff et al. \(2018\)](#) develops a two-component model based on the thrust required to balance the drone weight and the parasite drag force. This model is used to assess the energy use and life cycle greenhouse gas (GHG) emissions for small drones with short ranges (4 km) delivering from warehouses. Results indicate that small drones are likely to provide lower lifecycle GHG emissions than conventional delivery trucks, but that benefits depend on the carbon intensity of electricity and the size of the drones.

[Liu et al. \(2017\)](#) provide a detailed three component drone energy model that includes power to maintain lift and overcome parasite drag, along with profile power to overcome the rotational drag of the propeller blades. Field tests in [Liu et al. \(2017\)](#) show that ascending takes 9.8% more power than hovering, and descending takes 8.5% less power than hovering (similar to results in [Di Franco and Buttazzo \(2015\)](#)). [Kirchstein \(2020\)](#) uses a similar component model, but in an idealized delivery process from a depot with separate energy calculations for takeoff and ascent, steady level flight, descent, hovering, and landing for delivery. The return trip is similar but without the payload. [Kirchstein \(2020\)](#) compares delivery with trucks versus drones for Berlin and shows that drone delivery often requires more energy. The results also highlight how wind and drone hovering increase energy consumption. Several other authors have developed component drone power and energy models similar to those above for problems involving drones in wireless communication networks (e.g., [Zeng and Zhang, 2017; Zeng, 2019; Wu, 2019](#)).

A third approach to modeling drone energy consumption is with regression based on field experiments, as in some articles previously mentioned. [Tseng et al. \(2017a, 2017b\)](#) present a nine-term nonlinear regression model for drone power consumption that includes horizontal and vertical speed and acceleration, as well as payload mass and wind speed.

Lastly, an alternative approach to estimate drone energy consumption is to use information provided by the manufacturer, where the energy consumption rate (in Joules/meter) is calculated as $\frac{\text{BatteryEnergy(inJ)}}{\text{FlightDistance(inm)}}$. Information on battery energy is usually, but not always, provided by the manufacturer (e.g., in Watt-hours or as a capacity and voltage). However, flight distance information is not reported by all manufacturers, nor is it provided in a consistent manner. Further, flight distance data, when available from manufacturer websites, may be the distance traveled under conditions that maximize that distance (e.g., at the speed, altitude and weather conditions that maximize distance) which may not reflect real-world drone operations. While one can calculate a flight distance as the product of speed and flight time using reported data for a flight time (or hovering time) and drone speed, we note that manufacturers often report maximum values (e.g., "Max Flight Time" and "Max Speed") or unspecified values, and that drone energy use depends strongly on speed. Thus, estimates of flight distances and energy consumption are likely to be inaccurate when based on flight times and speeds reported by manufacturers that are either maximum values or from unknown operating conditions. Ideally, there would be standard energy efficiency data provided by manufacturers from an independent or governmental source (e.g., as with energy efficiency measures for automobiles or appliances). See Appendix A for further discussion of estimating Epm based on manufacturer data when

Table 1

Summary of key features of the theoretical and regression models of drone energy consumption in the literature.

Reference	Thrust Assumption for Flight			Travel Components			Wind	Avionics	Empty return	Theoretical Model	Regression Model	Field Tests
	i T = W/r	ii T = W	iii T = W + D + L	Horizontal	Hover	Vertical						
D'Andrea (2014)	x			x			x	x		x		
Figliozzi (2017)	x			x					x	x		
Dorling et al. (2017)		x		x	x	x				x	x	x
Tseng et al. (2017a)				x	x	x	x	x			x	x
Tseng et al. (2017b)				x	x	x	x	x			x	x
Liu et al. (2017)			x	x	x	x	x			x		x
Lohn (2017)	x			x	x	x	x	x	x	x		
Xu (2017)	x			x	x	x	x	x	x	x		
Stolaroff et al. (2018)			x	x			x		x	x		x
Troudi et al. (2018)	x			x				x		x		
Jeong et al. (2019)		x		x						x	x	
Kirchstein (2020)			x	x	x	x	x	x	x	x		

Notes for energy consumption adjustments: Including avionics increases energy consumption, while modeling empty returns decreases energy consumption (the return trip has lower weight). Wind is often modeled as increasing drone energy consumption (e.g., D'Andrea (2014)), though more detailed analyses show the energy use may increase or decrease depending on wind speed and direction and drone type (e.g., headwinds may increase lift, thereby reducing the power requirements; see Kirchstein (2020) for a detailed analysis of wind effects).

flight distance information is provided.

Murray and Raj (2020) design truck-drone tandem delivery routes with a three-phase heuristic and is the only reference to consider multiple drone energy models, including the model of Liu et al. (2017), a simple regression model that is linear in payload, and other models with a fixed distance or time limit (essentially modeling energy consumption as a linear function of drone travel distance or time). Relevant findings for our study were that (i) the different energy models can produce very different routes, with several energy models leading to the creation of energy infeasible drone routes, and (ii) it is important to include the energy consumed outside the steady level flight portion of a delivery trip (e.g., for launch, retrieval and delivery), especially for any hovering needed for coordination with a truck or other drones prior to landing.

Table 1 provides a categorization of 12 key drone energy consumption models in the literature. Column 1 identifies the reference for the model. Columns 2–4 identify the assumption regarding thrust for steady horizontal flight and show the three groupings of models as discussed earlier. Columns 5–7 reflect the scope of drone travel included in the model for the different drone flight segments (horizontal flight, hover, and vertical flight including takeoff/landing). Note that all 12 models for drone delivery include energy use for horizontal travel, but only half include energy use for hovering and vertical travel. Columns 8–10 indicate adjustments to the modeled energy consumption for wind, avionics, and empty returns. Note that only 2 models do not include at least one of the three adjustments in columns 8–10, although these adjustments are often removed from the models in the reported computational results. Columns 11–12 indicate the type of model provided (either theoretical based on modeling thrust as in columns 2–4, or regression models) and show that 10 articles provide theoretical models, while 3 papers present regression models. Column 13 indicates the five references that include field tests with a drone, often used for setting parameter values in the model.

3. Theoretical models for energy consumption

In this section, we discuss the theoretical and regression models for energy consumption of each of the 12 articles in Table 1. We first introduce in sub-section 3.1 a unified notation to facilitate comparison of all models. We then discuss in sub-section 3.2 the models that use an integrated approach (with the lift-to-drag ratio), followed in sub-section 3.3 by the models that use a more complex component approach. For consistency we report all energy values in Joules (1 Joule = 1 Watt-second) and energy consumption rate for steady level flight (E_{pm}) in Joules/meter.

Table 2

Unified notation for drone energy models.

ρ	= air density [kg/m ³] (e.g., 1.225 kg/m ³ at 15° C at sea level)
g	= acceleration of gravity [m/s ²]
v_i	= induced speed [m/s] (the change in the speed of the air after it flows through an object)
v_a	= airspeed [m/s] (speed of drone relative to the air)
φ	= ratio of headwind to airspeed [unitless]
v	= drone ground speed [m/s], so $v = (1 - \varphi)v_a$
d	= drone one-way travel distance for a single delivery trip [m]
r	= lift-to-drag ratio [unitless]
η	= battery and motor power transfer efficiency (from battery to propeller) [unitless]
η_c	= battery charging efficiency [unitless]
k	= index of the drone components: drone body = 1; drone battery = 2; payload (package) = 3
C_{D_k}	= drag coefficient of drone component k [unitless]
A_k	= projected area of drone component k [m ²]
m_k	= mass of drone component k [kg]
γ	= maximum depth of discharge of the battery [unitless]
s_{bat}	= specific energy of the battery (energy capacity per kg) [J/kg]
f	= safety factor to reserve energy in the battery for unusual conditions [unitless]
R	= maximum one-way distance of drone travel per battery charge [m]
P	= power required to maintain a steady drone flight [Watt = J/s]
P_{avio}	= power required for all avionics on the drone (independent of drone motion) [Watt = J/s]
n	= number of rotors for a rotorcopter drone [rotors]
N	= number of blades in one rotor for a rotorcopter drone [blades]
c	= blade chord length [m]
c_d	= blade drag coefficient [unitless]
ς	= area of the spinning blade disc of one rotor [m ²]
α	= drone angle of attack [radians]
Epm	= energy required for steady drone flight per unit distance [J/m]

3.1. Unified notation

Different authors use different notation for the same concepts, so to facilitate understanding and comparison of the models, we employ the unified notation shown in Table 2 (m = meter; s = seconds; J = joules). We classify the drone physical components into three categories: (i) drone body (including the airframe, propellers, motors, sensors, GPS, avionics, and a camera if used), (ii) drone battery, and (iii) payload (package). Thus, the drone body includes everything except the battery and package. Two areas that sometimes cause confusion in the literature concern the range and the battery usage. We use R to denote the maximum distance that a drone can travel in one direction and still be able to return to the depot, i.e., half of the round-trip distance for an out-and-back delivery. We define η as the power transfer efficiency, which is the energy loss from battery to the propeller. Figliozzi (2017) uses an overall power transfer efficiency, which also includes the energy loss from charging to battery, to denote power transfer efficiency. This definition will result in a value a little smaller than η , as charging is not 100% efficient.

3.2. Energy models using integrated approaches

The seminal integrated model is provided by D'Andrea (2014) and is based on the ratio of lift-to-drag, r . This is a simple formula for calculating the power consumption (in kJ/s) required for the drone to maintain steady flight and operate on-board electronics in terms of the drone total mass, its speed, the lift-to-drag ratio, and the battery's power transfer efficiency. The derivation of this model is given in Appendix B. With no wind, D'Andrea (2014) provides the expression

$$P = \frac{\sum_{k=1}^3 m_k v}{370 r \eta} + P_{avio} \quad (1)$$

where the constant 370 ($370 = 3600/9.8$) allows velocities to be expressed in km/h rather than meters per second. In this paper, we express all speeds in m/s, so the equivalent expression is

$$P = \frac{(\sum_{k=1}^3 m_k) g v_a}{r \eta} + P_{avio} \quad (2)$$

The energy consumed for steady flight over a distance d is the power multiplied by the travel time d/v , so the energy per meter of travel (Epm) is the power divided by the speed

$$Epm = \frac{P}{v_a} \quad (3)$$

So, for D'Andrea (2014) including possible headwinds as indicated via the unitless factor $\varphi = \frac{\text{headwindspeed}}{\text{droneairspeed}}$, we have

$$Epm = \frac{1}{1 - \varphi} \left(\frac{g \sum_{k=1}^3 m_k}{r\eta} + \frac{P_{avio}}{v_a} \right) \quad (4)$$

One example in D'Andrea (2014) with no headwind ($\varphi = 0$), uses $m_1 = m_2 = m_3 = 2$ kg, $v_a = 12.5$ m/s (45 km/hr), $r = 3$, $\eta = 0.5$ and $P_{avio} = 100$ J/s. This requires total power of $P = 590$ J/s, or $Epm = 47.2$ J/m. With an 8.33 m/s headwind, the Epm triples to 141.6 J/m. Troudi (2018) uses the same model as described in eq.(2) with a MD4-1000 drone with a battery of 1,040,400 Joules (or 289Wh), $m_3 = 1.2$ kg, $m_1 + m_2 + m_3 = 5.55$ kg, and $v_a = 13$ m/s.

Figliozzi (2017) extends the basic model from D'Andrea (2014) for “steady flight” to consider empty returns. This model does not include power for avionics ($P_{avio} = 0$), models the lift-to-drag ratio as dependent on speed (with $r(v)$), and provides a unitless parameter for battery recharging efficiency (η_r). The overall energy consumption rate per unit distance (with no headwind) is given as

$$Epm = \frac{1}{2} \left(\frac{g \sum_{k=1}^3 m_k}{r(v)\eta_r} + \frac{g \sum_{k=1}^2 m_k}{r(v)\eta_r} \right) \quad (5)$$

The first term in parentheses is the energy consumption rate with a package, and the second term is the energy consumption rate without a package on the return trip.

The parameters used by Figliozzi (2017) are for a MicroDrones MD4-3000 and are as follows: $m_1 + m_2 = 10.1$ kg, $m_3 = 5$ kg, and the drone has a range of 36 km, though the paper suggests 70% of this, or 25 km, as the maximum range to provide a safety margin and allow for “unknown factors that can increase energy consumption such as headwinds”. The lift-to-drag ratio $r(v)$ is not specified; however the article reports an Epm value of 77.8 J/m “calculated utilizing manufacturer information” and an overall power efficiency product of $\eta_r = 0.9 \times 0.73 = 0.66$. From these values, the lift-to-drag ratio can be calculated in reverse as $r = 2.4$. This article uses a battery of 2.8 million Joules (or 777Wh).

The energy model used in the RAND Corporation studies (Lohn, 2017; Xu, 2017; Gulden, 2017) is based on D'Andrea (2014) and described in Lohn (2017). Lohn (2017) reports that for truck and drone delivery serving a city the size of Los Angeles (1500 km²) from a centrally located drone depot, the average energy use rate is $Epm = 112.5$ J/m (assuming uniformly distributed deliveries in a circular city). Xu (2017) models a complete drone delivery mission includes ascending to 150 m of altitude, flying level at 22.2 m/s (80 km/hr) into a 2.8 m/s (10 km/hr) headwind and then descending to the delivery site, with 30 s of hovering. The return trip is similar but without the payload and also includes flying into a 2.8 m/s (or 10 km/hr) headwind. Results are provided for a baseline multicopter design based on an Amazon Prime Air VTOL drone with 8 lift rotors and 2 cruise motors and the lift-to-drag ratio $r = 3$. For this baseline hybrid drone (weighing 55 lbs with a 20 mile range), Xu (2017) provides an energy consumption of 5.4 MJ for a 2.3 kg payload flying 32.2 km, resulting in $Epm = 168.0$ J/m.

3.3. Energy models using component approaches

Dorling et al. (2017) provides a model for drone energy consumption based on hovering only (with the assumption that this is approximately equal to the energy for drone travel). During hover, the airspeed is zero, and the thrust balances the weight force, so

$$T = g \sum_{k=1}^3 m_k \quad (6)$$

Based on helicopter theory (e.g., Leishman (2002)), the power required for the drone to hover is

$$P = \frac{T^{3/2}}{\sqrt{2n\rho\zeta}} = \frac{(g \sum_{k=1}^3 m_k)^{3/2}}{\sqrt{2n\rho\zeta}} \quad (7)$$

where n is the number of rotors, and ζ is the area of the spinning blade disc of one rotor.

The Epm can then be calculated as

$$Epm = P/v_a = \frac{(g \sum_{k=1}^3 m_k)^{3/2}}{v_a \sqrt{2n\rho\zeta}} \quad (8)$$

This does not include power for avionics or an adjustment for wind or power transfer efficiency η .

Field tests conducted with the Hexa-B drone with no payload showed that hovering consumed 2.3% more power than steady flight at 6 m/s, and 4.7% more power than repeated altitude changes (vertical flight) from 0 to 25 m. Field test data while hovering with small payloads of 0.5–1.5 kg provided very different, and much larger power consumption levels than the theoretical model of eq.(7) (4.5 to 5 times larger!).

The authors also approximate the hover power consumption as a linear function of the

battery and payload weight for an ArduCopter Hexa-B drone with $n = 6$, $\zeta = 0.2$ m², and a frame weight $m_1 = 1.5$ kg as

$$P = \beta_1(m_2 + m_3) + \beta_0 \quad (9)$$

where β_1 is the power consumed per kilogram of battery and package weight, and β_0 is the power required to keep the drone frame of mass m_1 in the air. Values of $\beta_1 = 46.7$ J/s-kg and $\beta_0 = 26.9$ J/s were generated from eq. (7) using linear regression with $m_2 + m_3$ ranging from 0 to 3 kg in increments of 0.001 kg. Regression parameters for eq. (9) based on the field tests with small payloads provide much larger values of β_1 and β_0 , to provide

$$Epm = P/v_a = (217(m_2 + m_3) + 185)/v_a \quad (10)$$

$$Epm = P/v_a = (171(m_2 + m_3) + 187)/v_a \quad (11)$$

for a large 14.8 V battery and a small 11.1 V battery, respectively.

Stolaroff et al. (2018) provide a component model based on the fundamental forces of flight (including forces of the weight, the parasite and induced drags) and also conduct field testing. For a hovering rotocopter, the same power model is provided as in eq.(7). But for forward flight, Stolaroff et al. (2018) present a model for thrust

$$T = W + D = g \sum_{k=1}^3 m_k + \frac{1}{2} \rho \sum_{k=1}^3 C_{D_k} A_k v_a^2 \quad (12)$$

where the first term reflects the total drone weight and the second term is the parasite drag force, which depends on the drag coefficient C_{D_k} and the projected area perpendicular to travel A_k of each drone component (airframe, battery and payload). Note that when the drone hovers, the airspeed equals zero ($v_a = 0$) and so $T = \sum_{k=1}^3 m_k g$, as in eq.(7).

With forward flight, or heavy wind, Stolaroff et al. (2018) use a power consumption formula adapted from Hoffmann et al. (2007)

$$P = \frac{T(v_a \sin \alpha + v_i)}{\eta} \quad (13)$$

where α is the angle of attack (i.e., the angle of the airspeed to the drone rotor) and v_i is the induced speed, which can be found by solving (numerically)

$$v_i = \frac{g \sum_{k=1}^3 m_k}{2n\rho\zeta \sqrt{(v_a \cos \alpha)^2 + (v_a \sin \alpha + v_i)^2}} \quad (14)$$

The angle of attack α is given by

$$\alpha = \tan^{-1} \left(\frac{\frac{1}{2} \rho \left(\sum_{k=1}^3 C_{D_k} A_k \right) v_a^2}{g \sum_{k=1}^3 m_k} \right) \quad (15)$$

For large values of α the drone may become unstable, so in practice α may be limited to maintain stable flight (Ai, 2019; DJI website, 2020). The overall energy per meter is then given as

$$Epm = \frac{T(v_a \sin \alpha + v_i)}{v_a \eta} \quad (16)$$

with α from eq.(15) and v_i being the solution of eq.(14). Empty returns are assumed.

Stolaroff et al. (2018) consider a small quadcopter of total mass 2.57 kg, including a 0.5 kg payload (3D Robotics' Iris), and a larger octocopter of total mass 24 kg, including a 7 kg payload (Turbo Ace's Infinity 9). Field measurements consisting of 1073 flight segments with the small quadcopter in moderate winds (up to 7 m/s at random orientation to the direction of travel) are used to set some parameter values, including the power transfer efficiency $\eta = 0.7$. See Stolaroff et al. (2018) for details and all parameter values.

Stolaroff et al. (2018) also model the maximum range of a drone assuming that the drone carries the payload in one direction only (i.e., $m_3 = 0$ on the return trip). This is expressed as a function of the battery mass (m_2), battery energy capacity per kg (s_{batt}), the depth of battery discharge (γ), a battery safety factor f (assumed to be 1.2), and the energy consumption rate carrying the payload Epm_{loaded} and on the empty return $Epm_{unloaded}$

$$R = \frac{m_2 s_{batt} \gamma}{(Epm_{loaded} + Epm_{unloaded})f} \quad (17)$$

Liu et al. (2017) provide a more comprehensive and detailed energy consumption model for small quadrotor UAVs. The model has three components: (i) induced power to maintain lift, which is a function of the thrust and the vertical drone speed, (ii) parasite power as in Stolaroff et al. (2018), and (iii) profile power to overcome the rotational drag encountered by propeller blades, which is a function of the thrust and the airspeed (see Rotaru and Todorov (2017) for related details on power requirements for helicopters).

The parasitic power is the same as in Stolaroff et al. (2018) and is given by $\sum_{k=1}^3 \frac{1}{2} \rho v_a^3 C_{D_k} A_k$, although the authors do not explicitly consider the drag forces for the battery and the payload. The other two power components depend on the thrust, which is given by

$$T = \sqrt{\left(g \sum_{k=1}^3 m_k - c_5 (v_a \cos \alpha)^2\right)^2 + \left(\frac{1}{2} \rho \left(\sum_{k=1}^3 C_{D_k} A_k\right) v_a^2\right)^2} \quad (18)$$

where the term $g \sum_{k=1}^3 m_k$ is the weight to be lifted, the term $c_5 (v_a \cos \alpha)^2$ reflects the lift generated from horizontal movements (likely to be especially important for hybrid drones with lifting surfaces), and the last term is again the parasitic drag. For a drone in steady level flight, the Epm , based on the total power from equation (14) in Liu et al. (2017), is

$$Epm = \frac{\kappa_1}{\sqrt{2n\rho\zeta}} \frac{T^{3/2}}{v_a} + \frac{1}{2} \rho \left(\sum_{k=1}^3 C_{D_k} A_k\right) v_a^2 + c_2 \frac{T^{3/2}}{v_a} \quad (19)$$

where the first term reflects the induced power, the second term reflects the parasitic power and the last term reflects the profile power, with parameter c_2 depending on the air density and details of the rotors (including the efficiency of converting rotor angular speed to thrust, the number of blades in each propeller, the blade chord width, drag coefficient, and length). Based on field testing, Liu et al. (2017) report that with a 1.43-kg 3DR IRIS + quadcopter, $\kappa_1 = 0.8554$ and $c_2 = 0.3177 \text{ (m/kg)}^{1/2}$. Note that including the profile power in the component model requires detailed decisions on the drone rotors and motors (for calculating c_2).

Liu et al. (2017) include some field tests, with the most involved having the drone with no payload ascend to 70 m, fly a horizontal rectangular loop for about 730 m, then descend and land back at the origin. Comparing the measured power and the estimated power from the proposed models shows that the model underestimated the energy consumed in ascending by 10.7%, underestimated the energy consumed in horizontal flight by 16.3%, and overestimated the energy consumed in descending by 2.2%. For the total flight, the model underpredicted the energy by 11.4%. Because the experiment was for a short flight, ascending and descending consumed 39% of the total energy; in contrast to a 10 km horizontal trip, ascent and descent would consume only 4.5% of the total energy. For the rectangular loop trip, i.e., the entire horizontal flights (of length 730 m), the modeled energy consumption is $1.23 \times 10^4 \text{ J}$ which equates to about $Epm = 17.3 \text{ J/m}$.

Kirschstein (2020) provides a component model originally from Langelaan et al. (2017) based on an idealized delivery process (like Xu (2017)) with takeoff and ascent at 45° to a cruising altitude (150 m), level flight, descent (at 45°) with hovering, then landing for delivery. The return is similar but without the payload. Like Liu et al. (2017) the model includes energy consumed for induced power, parasite power, and profile power; but the model also includes power for climbing and avionics, and adjustments for power losses due to the electric motor, transmission and charging efficiencies. For steady flight, Kirschstein (2020) uses a thrust of

$$T = \sqrt{\left(g \sum_{k=1}^3 m_k\right)^2 + \left(\frac{1}{2} \rho \left(\sum_{k=1}^3 C_{D_k} A_k\right) v_a^2\right)^2} + \rho \left(\sum_{k=1}^3 C_{D_k} A_k\right) v_a^2 \text{mg} \sin \theta \quad (20)$$

where the flight angle θ allows ascending and descending to be modeled. For hovering (i.e., $v_a = 0$), the thrust reduces to $g \sum_{k=1}^3 m_k$. Kirschstein's general energy model is detailed in Appendix C and the Epm for steady level flight (i.e., $\theta = 0$) with no wind can be written as

$$Epm = \frac{1}{\eta} \left(\frac{\kappa T w}{v_a} + \frac{1}{2} \rho \left(\sum_{k=1}^3 C_{D_k} A_k\right) v_a^2 + \frac{\kappa_2 (g \sum_{k=1}^3 m_k)^{1.5}}{v_a} + \kappa_3 \left(g \sum_{k=1}^3 m_k\right)^{0.5} v_a \right) + \frac{P_{avio}}{\eta_c v_a} \quad (21)$$

The first term in the Epm is for induced power with κ being the “lifting power markup” and w the “downwash coefficient” (See Kirschstein (2020) for details). The second term in the Epm is for parasite drag (as in Liu et al. (2017) and Stolaroff et al. (2018)). The third and fourth terms are for profile power where constants κ_2 and κ_3 reflect details of the rotors and environment (as in Liu et al. (2017)). The last term in the Epm is for avionics.

Kirschstein (2020) provides analyses for delivery in an urban region (comparing drones with diesel and electric trucks) with large octocopter drones ($m_1 + m_2 = 12 \text{ kg}$) that carry a 2.5 kg payload ($m_3 = 2.5 \text{ kg}$), travel at 22.2 m/s, have a flight radius of 9 km (and the trips include 5 min of hovering), power transfer efficiency $\eta = 0.73$, battery charging efficiency $\eta_c = 0.9$ and $P_{avio} = 100 \text{ J/s}$ (as in D'Andrea (2014)). For this drone, the constants in eq. (20) are: $\kappa = 1.15$, $\kappa_2 = 0.502$ and $\kappa_3 = 0.118$; and the loaded $Epm = 131.5 \text{ J/m}$.

Tseng et al. (2017a, 2017b) present a nine-term nonlinear regression model for drone power use that includes horizontal and vertical speeds and acceleration, as well as payload mass and wind speed, and provide a good fit to the reported field data. The regression model for steady flight with no wind reduces to $P = \beta_1 v_a + \beta_7 m_2 + \beta_9$. They collected data to estimate the parameters based on test drone flights using a 3DR Solo drone (weighing 2 kg) with small payloads of 0 kg, 0.25 kg, and 0.5 kg, and a DJI Matrice 100 drone (weighing 2.8 kg) with small payloads of 0 kg, 0.3 kg, and 0.6 kg. The regression equation for Epm in Tseng et al. (2017b) and Tseng (2020) for steady flight with speeds up to 5 m/s and no wind for the larger DJI Matrice 100 drone is

$$Epm = \frac{P}{v_a} = -1.526 + \frac{0.220 m_2 + 433.9}{v_a} \quad (22)$$

and for the smaller 3DR Solo drone it is

$$Epm = -2.595 + \frac{0.197 m_2 + 251.7}{v_a} \quad (23)$$

Table 3a

Key drone energy models.

Color	ID	Reference	Model Type ¹	Drone Type (index)	$m_1 + m_2$ (kg)	Airspeed v_a (m/s)	Equation for E_{pm}	Field tests
Blue	L-4-Mi	Liu et al. (2017)	C	Quadcopter (4)	1.46	8–23	(18–19)	X
Blue	S-4-Mi	Stolaroff et al. (2018)	C	Quadcopter (4)	2.07	10	(12–16)	X
Blue	DH-6-Mi	Dorling et al. (2017)	C	Hexacopter (6)	2	6 ²	(8)	
Blue	DR-6-Mi	Dorling et al. (2017)	R	Hexacopter (6)	2	6	(10)	X
Blue	T1-4-Mi	Tseng et al. (2017b)	R	Quadcopter (4)	2	0–5	(22)	X
Blue	T2-4-Mi	Tseng et al. (2017b)	R	Quadcopter (4)	2.8	0–5	(23)	X
Blue	A-G-Mi	D'Andrea (2014)	I	General (G)	4	12.5	(4)	
Orange	F-4-L	Figliozzi (2017)	I	Quadcopter (4)	10.1	unknown	(5)	
Orange	K-8-L	Kirchstein (2020)	C	Octocopter (8)	12	22.2	(20–21)	
Red	S-8-L	Stolaroff et al. (2018)	C	Octocopter (8)	17	10	(12–16)	X
Red	X-H-L	Xu (2017)	I	Hybrid-current (H)	25	22.2	Not available ³	

¹ C = component model; R = regression model; I = integrated model; ²the setting in Dorling et al. (2017) uses a drone speed of 6 m/s although the energy equation (8) does not include speed; ³based on D'Andrea (2014)

4. Analysis and results

In this section, we compare results for the key drone energy models described in Section 3 to explore the interrelated aspects of payload mass, speed, energy consumption per meter and maximum drone range. The energy use and drone range are outputs determined by (i) specifics of the drone and the battery modeled, (ii) operations, specifically speed and payload mass, and (iii) energy adjustments, such as, wind, avionics, and energy loss.

Table 3a summarizes the 11 baseline drone energy models examined in Section 3, arranged in order of increasing drone weight. Columns 1–2 provide an identifying color and ID used in later figures. The first part of the ID (1 or 2 characters) indicates the reference and model, where DH and DR respectively indicate the theoretical hovering and the regression models in Dorling et al. (2017), and T1 and T2 indicate the two regression models from Tseng et al. (2017b). The second part of the ID is the number of rotors for the drone or the drone type. The third part of the ID indicates the drone weight based on the Table 4 classification of Hassanalian and Abdelkefi (2017), with “Mi” for micro drones (≤ 5 kg) and “L” for light drones (5–50 kg). Note that the majority of the baseline models are for small micro rotocopter drones with small payloads. Column 3 identifies the reference. Column 4 describes the model type (integrated, component or regression) as discussed in Section 3. Columns 5–7 provide information on the drone type, mass and airspeed for the baseline setting in the reference. Column 8 is the equation number in this paper for E_{pm} . Column 9 indicates if parameter values used in the equation were based on field experiments.

Table 3b provides further information with Columns 1–2 showing the ID and reference. Column 3 indicates the E_{pm} value, with ranges shown for the regression models. Columns 4–7 provide the takeoff weight, drone body weight, battery weight, and payload. Columns 8–9 provide the battery energy (in KJ) and the battery energy density (battery energy/battery weight).

4.1. E_{pm} Results from the literature

Fig. 3 shows a graph of E_{pm} versus payload mass for the models in Tables 3a and 3b using the parameter values and operational setting in the reference. There are three short lines for the regression models (DR-6-Mi, T1-4-Mi, T2-4-Mi), and one point each for the other eight models. We emphasize that these data are for the specific drone and the operational setting considered in each reference (i. e., the drone size, payload and speed shown in Table 3a) in steady level flight with no wind or avionics, with the exception of Xu (2017) which includes a complete flight profile (ascent, level flight, hovering and descent on the forward and return trip). Xu (2017) does not provide a model that allows the steady level flight portion of the trip to be extracted (although the steady level flight for the baseline model in Xu (2017) would be 98.8 J/m if steady level flight accounted for the same percentage of the total energy as in Kirchstein (2020), which uses a similar trip profile).

Fig. 3 documents the wide range of energy consumption values per unit of distance reported in the literature for similar delivery missions, with payloads that range from 0 to 7 kg. As expected, the general trend is evident that larger (heavier) drones and payloads have larger E_{pm} values; however, the range of E_{pm} values varies substantially, both across the payloads and for similar payloads. The models for micro drones with payloads ≤ 1 kg (and airspeeds ≤ 6 m/s) cluster in the lower left of Fig. 3 with E_{pm} values ranging from 15.7 J/m to 107.3 J/m. The E_{pm} for drones with slightly greater payloads of 1.15–2.5 kg (and airspeeds of 12–23 m/s) range from 39.2 J/m to 168.0 J/m, though as noted above X-H-L includes a complete delivery profile. Also note that Fig. 3 shows the average payload for the three models with empty returns, X-H-L, K-8-L and F-4-L, where the outbound payload is 2.3, 2.0 and 5 kg, respectively. The S-8-L model for the drone with the largest payload (7 kg) provides by far the largest E_{pm} of 436.7 J/m, which is more than 5 times larger than the energy use for the drone with the next largest payload (the F-4-L model with an outbound 5 kg payload). While Fig. 3 displays E_{pm} values from the models or field tests for the baseline settings described in the corresponding articles, these models could be used for other delivery settings, such as delivering lighter payloads. For example, using the Stolaroff et al. (2018) heavy drone S-8-L model with a small 2 kg payload would reduce the E_{pm} to 321.8 J/m, which is still well above the values reported for the other models with similar payloads.

Key findings revealed by Fig. 3 are: (i) the energy consumption rates reported in the literature for steady level flight vary

Table 3b

Weight and energy information for key drone energy models.

ID	Reference	Epm (J/m)	Takeoff Weight (kg)	Body Weight (kg)	Battery Weight (kg)	Payload (kg)	Battery Energy (KJ)	Battery Energy Density (KJ/kg)
L-4-Mi	Liu et al. (2017)	17.3	1.46	1.14	0.32	0	204	637
S-4-Mi	Stolaroff et al. (2018)	47.4	2.57	1.07	1	0.5	540	540
DH-6-Mi	Dorling et al. (2017)	15.7	3.5	1.5	0.5	1	325	650
DR-6-Mi	Dorling et al. (2017)	48.9–85.1	3.5	1.5	0.5	0–1	325	650
T1-4-Mi	Tseng et al. (2017a)	47.7–67.4	2.5	1.5	0.5	0–0.5	277	554
T2-4-Mi	Tseng et al. (2017a)	85.3–107.3	3.4	2.2	0.6	0–0.6	360	599
A-G-Mi	D'Andrea (2014)	39.2	6	2	2	2	900	450
F-4-L	Figliozzi (2017)	77.8	12.6	6	4.1	2.5 ¹	2797	682
K-8-L	Kirchstein (2020)	127.2	13.25	2	10	1.25 ¹	5400	540
S-8-L	Stolaroff et al. (2018)	436.7	24	7	10	7	5400	540
X-H-L	Xu et al. (2017)	168.0	23.85	13.3	8.3	1.15 ¹	5806	702

¹ Average payload as the model assumes an empty return; the forward trip carries twice the payload in the table and the return trip has no payload.

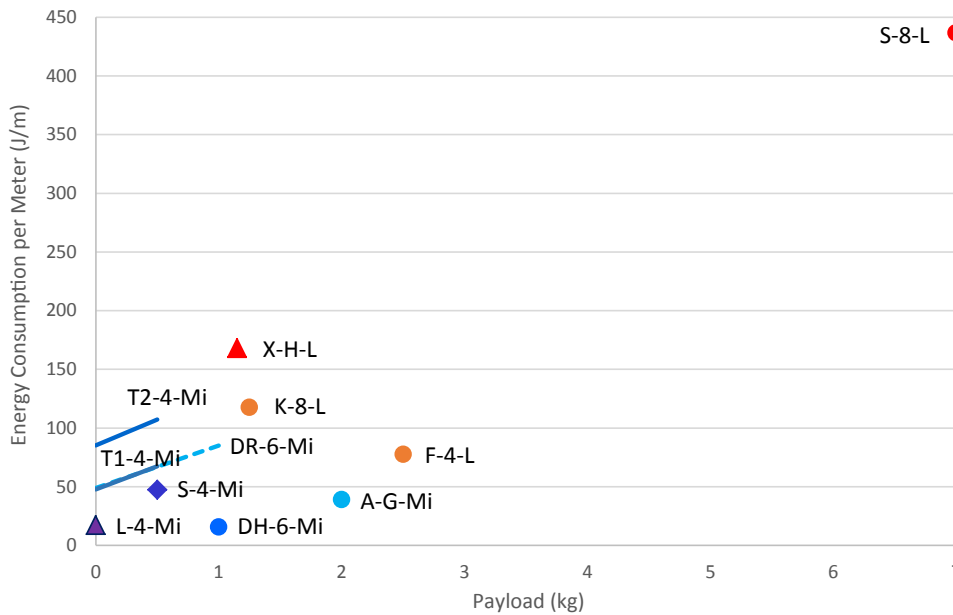


Fig. 3. *Epm* results from baseline models in Tables 3a and 3b.

substantially with the payload, from under 20 J/m to over 400 J/m for payloads up to 7 kg; and (ii) with similar payloads the energy consumption models differ by factors of several hundred percent. Other drone research that does not employ an explicit energy model, but rather uses a fixed value for *Epm*, such as Ferrandez et al. (2016) (46.1 J/m), Moore (2019) (223.0 J/m) or Goodchild and Toy (23.0–223.1 J/m) also show a very wide range of energy consumption rates. These findings document the lack of consensus regarding the energy efficiency of drones, and have important implications when drawing conclusions from modeling drone operations.

Four of the references in Tables 3a and 3b identify and consider adjustments to the baseline energy consumption rates due to factors such as wind and weather, avionics, flight profiles and future advances in technology. Table 4 summarizes the adjusted *Epm* values for the four models. The first row in each grouping for each reference reflects the baseline setting (no avionics, or wind), with the other rows showing adjustments for various additional conditions. The format of Table 4 is similar to Tables 3a and 3b, but with column 3 added to show the additional modeling considerations and column 9 showing the percentage change from the baseline.

The first three rows for the D'Andrea (2014) model show the moderate increase (20%) in *Epm* due to including avionics, and the very large increase (261%) from including a strong wind (30 km/hr). The next three rows show the assumed 50% increase in *Epm* in Figliozzi (2017) due to “adverse weather”, and the assumed 50% reduction for a future advanced drone. The next four rows for Kirchstein (2020) show a very small increase (3.4%) in *Epm* from including avionics, a very large increase (129%) due to including strong winds of 45 km/hr, and a substantial increase (70%) by including both moderate winds (10 km/hr) and energy use for a complete flight profile. The final three rows for Xu (2017) show the very large increase in *Epm* of 213% from using a larger drone with a 9.2 kg payload (and a shorter range due to using a fixed battery size), and a hypothetical large reduction in *Epm* of 83% from future advances in drone designs (especially in improving the lift-to-drag ratio).

Table 4 demonstrates that incorporating more realistic operating conditions (e.g., especially high winds) and a complete delivery profile (beyond just steady level flight) increases the *Epm* estimates substantially, in several cases over 100%. This suggests that using *Epm* values for steady level flight may significantly underestimate the total energy consumption of a drone trip. On the other hand, two researchers suggest that future developments in drone design have the potential to substantially reduce the energy requirements for drone travel. While future drone technology developments are difficult to predict, the large impact of strong winds on energy consumption is an area that needs further research, especially with field measurements in real operating environments. Some drone applications may allow flexible timing (such as scheduling inspections) to avoid wind and bad weather, while other applications (home delivery or surveillance) may have very limited time flexibility.

In summary, the published literature on drone energy consumption shows a strong lack of consensus (and standards) on the appropriate energy consumption rates, even for similar drone sizes and operations. The lack of consensus shown in Fig. 3 is further amplified by the results in Table 4 from including additional practical considerations. The energy consumption estimates of drones vary substantially from under 20 J/m to over 500 J/m. Because these differences are partly due to the different drones (e.g., different sizes, motor and rotor details, etc.) and settings (speed, payload, wind, etc.) employed, and partly due to the different model structures, in the following section we compare the models in a common setting with a common set of data reflecting two prototypical drones.

4.2. Results for different modeling approaches with a common drone and operational setting

To assess the drone energy consumption models from the literature on a common basis, we evaluate the five fundamental models

Table 4

Key drone energy models with additional conditions.

ID	Reference	Additional conditions to baseline	$m_1 + m_2$ (kg)	Payload ¹ (kg)	Airspeed v_a (m/s)	Range ² (km)	Epm(J/m)	% Epm increase to baseline
A-G-Mi	D'Andrea (2014)	Baseline	4	2	12.5	10	39.2	–
		Avionics					47.2	20.0%
		Wind (30 km/h), avionics					141.6	261%
F-4-L	Figliozzi (2017)	Baseline	10.1	2.5 ¹	unknown	25	77.8	–
		Adverse weather					116.7	50.0%
		Advanced drone					38.9	–50.0%
K-8-L	Kirchstein (2020)	Baseline	12	1.25 ¹	22.2	9	127.2	–
		Avionics					131.5	3.38%
		Wind (45 km/h), avionics					300.6	129%
		A complete flight profile with wind (10 km/h) & avionics					216.4	70.1%
X-H-L	Xu (2017)	Baseline: a complete flight profile with wind (10 km/h) & avionics	25.0	1.15 ¹	22.2	16.1	168.0	–
		Large drone	70.8	4.6 ¹		12	525.0	213%
		Advanced drone	5.2	1.15 ¹		16.1	29.0	–82.7%

¹ Average payload as the model assumes an empty return; the forward trip carries twice the payload in the table and the return trip has no payload.² Range = half of the round-trip distance.

Table 5

Five fundamental models for drone energy consumption of steady level flight.

Model	Key Reference	E_{pm}
LD	D'Andrea (2014)	$\frac{\sum_{k=1}^3 m_k g}{\eta \eta}$
RH	Dorling et al. (2017)	$\left(\sum_{k=1}^3 m_k g\right)^{3/2}$
R2	Stolaroff et al. (2018)	$\frac{\eta v_a \sqrt{2 \eta \rho \zeta}}{T(v_a \sin \alpha + v_i)}$ where $T = \sum_{k=1}^3 m_k g + \frac{1}{2} \rho \sum_{k=1}^3 C_{D_k} A_k v_a^2$ and v_i is found from solving equations (14–15)
R3	Kirchstein (2020)	$\frac{1}{\eta} \left(\frac{\kappa T w}{v_a} + \frac{1}{2} \rho \sum_{k=1}^3 C_{D_k} A_k v_a^2 + \frac{\kappa_2 \left(\sum_{k=1}^3 m_k g\right)^{1.5}}{v_a} + \kappa_3 \left(\sum_{k=1}^3 m_k g\right)^{0.5} v_a \right)$ where $T = \sqrt{\left(\sum_{k=1}^3 m_k g\right)^2 + \left(\frac{1}{2} \rho \sum_{k=1}^3 C_{D_k} A_k v_a^2\right)^2}$ and w is from solving equation (B2)
LR	Tseng (2017b)	$-2.595 + \frac{0.197 m_2 + 251.7}{v_a}$ (for small payloads and speeds less than 5 m/s)

Table 6

Parameter values used in drone energy use models in a common setting.

Panel 1. Parameter values that are independent of drone type			
Term	Symbol	Value	
Air density [kg/m ³]	ρ	1.225	
Acceleration of gravity [m/s ²]	g	9.807	
Ratio of headwind to airspeed [unitless]	φ	0	
Empty return (1 = yes; 0 = no)	ϕ	1	
Specific energy of the battery [J/kg]	s_{batt}	540,000	
Battery power transfer efficiency (from battery to propeller) [unitless]	η	0.7	
Safety factor to reserve energy in the battery for unusual conditions [unitless]	f	1.2	
Maximum depth of discharge of the battery [unitless]	γ	0.5	
Panel 2. Parameter values that depend on drone type			
Term	Symbol	Small Drone	Large Drone
Number of blades in one rotor [unitless]	N	4	3
Blade chord length [m]	c	0.0157	0.1
Blade lift coefficient [unitless]	c_l	0.271	0.4
Blade drag coefficient (depends on the airfoil)	c_d	0.012	0.075
Number of rotors [unitless]	n	4	8
Spinning area of one rotor [m ²]	ς	0.0507	0.027
Mass of drone body [kg]	m_1	1.07	7
Mass of battery [kg]	m_2	1	10
Mass of payload [kg]	m_3	0.5	7
Projected area of drone body [m ²]	A_1	0.0599	0.224
Projected area of battery [m ²]	A_2	0.0037	0.015
Projected area of payload [m ²]	A_3	0.0135	0.0929
Drag coefficient of drone body [unitless]	C_{D_1}	1.49	1.49
Drag coefficient of battery [unitless]	C_{D_2}	1	1
Drag coefficient of payload [unitless]	C_{D_3}	2.2	2.2
Lift-to-drag ratio [unitless]	r	3	3
Power required for avionics [Watt=J/s]	P_{avio}	0	0
Factor for induced power [unitless]	κ	1	1
Factor for profile power (m/kg) ^{1/2}	κ_2	0.790	0.683
Factor for profile power associated with speed (m/kg) ^{-1/2}	κ_3	0.0042	0.0868
Factor for parasite power with payload (kg/m)	κ_4	0.075	0.339
Factor for parasite power without payload (kg/m)	κ_4	0.057	0.214

discussed in Section 3 using common drone design parameters and a common operational setting, where the payload and speed are allowed to vary. The five models are identified in Table 5, along with the reference and their equation(s) for E_{pm} .

The first model, denoted LD, is the integrated model with a lift-to-drag ratio, based on eq. (4) from D'Andrea (2014). The second model, denoted RH, includes the energy use for rotorcopter hovering only, as in eq.(8) from Dorling et al (2017). The third model,

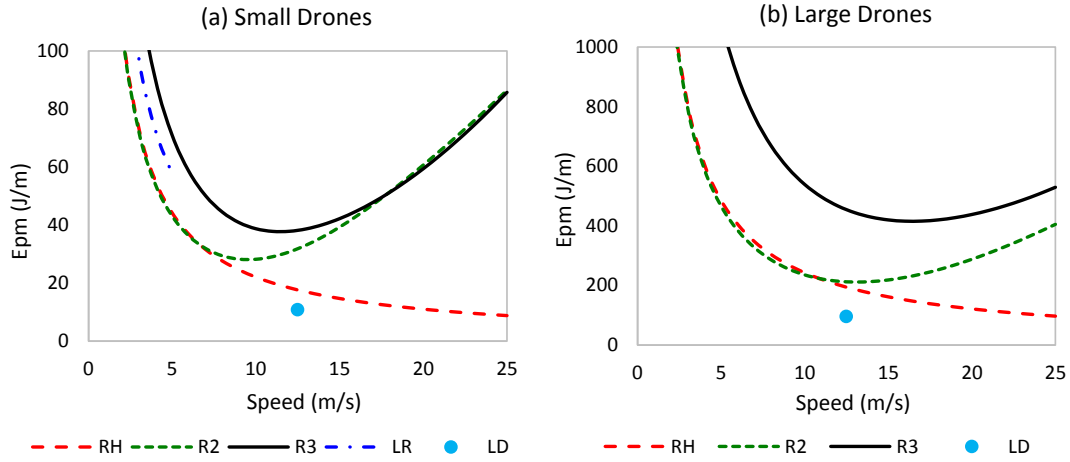


Fig. 4. Energy consumption rate versus airspeed for small and large drones.

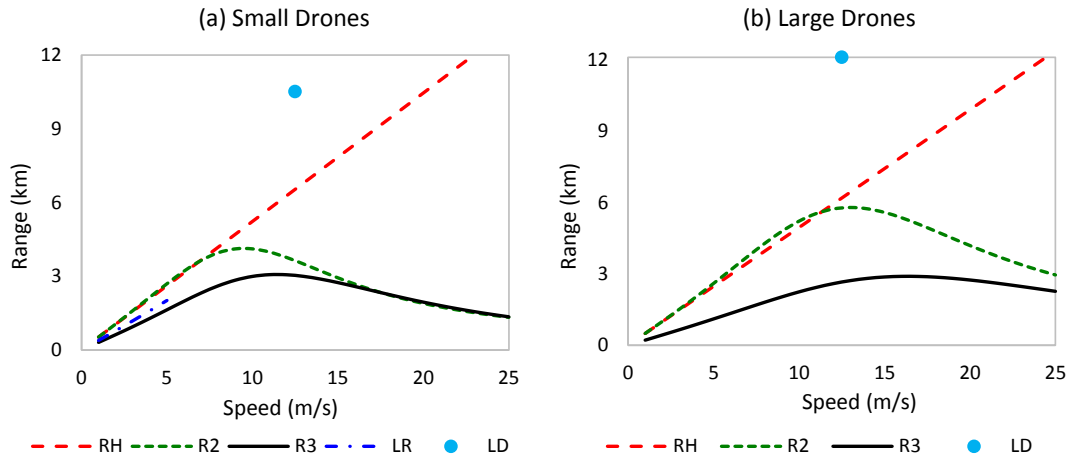


Fig. 5. Maximum flight range versus airspeed for small and large drones.

denoted R2, includes two rotorcraft energy components to overcome the induced drag and parasite drag, as in [Stolaroff et al. \(2018\)](#). The fourth model, denoted R3, includes three rotorcraft energy components, as it adds the profile drag to the induced and parasite drag, as in [Kirchstein \(2020\)](#). The final model, denoted LR, is the regression model from Tseng (2017b) that provides E_{pm} as a function of payload mass and airspeed. Because the LR regression model in [Tseng et al. \(2017b\)](#) was developed specifically for a small drone, we do not include that model for large drones.

To compare the different models on a common basis we use the set of parameters shown in [Table 6](#) (based mainly on data in [Stolaroff et al. \(2018\)](#)). Environmental parameters (e.g., gravity, air density, etc.) independent of drone design are fixed to the common values shown in Panel 1 of [Table 6](#). To model small and large drones, we use the associated parameters values shown in Panel 2 of [Table 6](#). The first set of parameters in column 3 of Panel 2 is associated with a small quadcopter drone capable of carrying a payload of 0.5 kg, such as might be used for medicine deliveries. The second set of parameters in column 4 of Panel 2 is associated with a larger octocopter drone capable of carrying a payload of 7 kg, such as might be used for home delivery of consumer goods. Note that the large drone battery has ten times the energy available as for the small drone. For all models, we include a common value for the power transfer efficiency of the drone, and we also assume empty returns so the E_{pm} values are the average of the loaded (with the payload) and unloaded (without the payload) E_{pm} values. We consider an environmental setting with no wind, and do not include energy consumption for avionics, vertical flight or hovering. We vary either the airspeed or payload mass to explore their effects on E_{pm} and range. Thus, by using the same drone specification and flight conditions, we can document the E_{pm} and flight range variabilities due to the different model structures and assumptions (not due to different input data). The performance measures of interest are E_{pm} and flight range. Range is calculated from eq. (17) based on the E_{pm} calculated using the equations in [Table 5](#).

4.2.1. The impact of speed on E_{pm} and range for small and large drones

We first consider the effect of airspeed on E_{pm} and range and identify the airspeeds that minimize E_{pm} . We note that the LD model is

Table 7Comparison of models at the *Epm* minimizing speed for R2 and R3.

	Model	<i>Epm</i> (J/m)	Speed (m/s)	Range (km)
Small Drone	LD	10.8	12.5 ¹	10.4
	RH	36.8	6 ¹	3.1
	R2	28.2	10	4.0
	R3	37.8	11	3.0
Large Drone	LD	95.7	12.5 ¹	11.8
	RH	404.8	6 ¹	2.8
	R2	211.6	13	5.3
	R3	415.8	16	2.7

¹ speed used by author in presenting the model

a special case because it does not depend on airspeed (see Table 5), so we report results with the LD model using the cruising airspeed of 12.5 m/s from D'Andrea (2014). A more sophisticated version of the LD model could use a speed dependent lift-to-drag ratio $r(v)$ to capture the impact of airspeed on *Epm*, but that is beyond the scope of this paper.

To analyze the effect of airspeed, we fix the payload at 0.5 kg for small drones and 7 kg for large drones and allow the airspeed to vary from 1 to 25 m/s. Results for energy consumption and flight range are shown in Figs. 4 and 5, respectively. The LR model in Fig. 4 (a) and 5(a) are shown only for speeds up to 5 m/s as that is the relevant speed range for the regression (Tseng, 2017b). The LD model is represented by a point at 12.5 m/s, the cruising speed used in D'Andrea (2014). For the other models, the *Epm* is a decreasing function of airspeed for low speeds, with R3 providing the highest *Epm* and RH and R2 providing nearly identical *Epm* values (R2 and RH are remarkably similar for speeds up to 8 m/s and 11 m/s for small and large drones, respectively). *Epm* is a convex function of airspeed for the component models R2 and R3, as overcoming the parasite drag at high speeds requires a large increase in energy.

The shapes of the curves for *Epm* are similar for both small and large drones, though the curves are less steep for large drones, and at higher speeds the *Epm* for model R3 remains above R2 with the large drone. For the RH, R2 and R3 models, the *Epm* is more sensitive to lower speeds than higher speeds; and while the models all have similar shapes for low speeds, they may produce very different values. For example, at 5 m/s the *Epm* values range from 43.1 J/m for R2 to 70.9 J/m for R3 for small drones (Fig. 4(a)) and from 467.5 to 485.8 for R2 and RH, to 1090.3 for R3 for large drones (Fig. 4(b)). The differing nature of the RH model (based on hovering) is clear in Fig. 4 as with higher speeds its *Epm* continues to decline as it is unable to capture the effect of increasing air resistance. The LD models provides an *Epm* that is generally lower than the other models, implying that the LD model with $r = 3$ represents a more efficient drone.

Fig. 5(a) and 5(b) show the drone flight range versus airspeed for the models for small and large drones, respectively. As with *Epm*, the range for the LD model does not change with speed, and is 10.4 km and 11.8 km for small and large drones, respectively, at 12.5 m/s. The range for the RH and LR models is linearly increasing in speed. The range for the R2 and R3 models follows a more complex pattern reflecting the convex *Epm* graphs. For small drones, the maximum range for the R2 and R3 models (which occurs with the energy-minimizing speed) is about 4 km and 3 km, respectively, less than half the range from the LD model. For large drones, the maximum range for the R2 and R3 models is about 5.7 km and 2.9 km, respectively.

We note that slower speeds quickly increase energy consumption (see Fig. 4), so the range for all models (except LD) falls quickly for slow drone speeds. For speeds below about 9 m/s for the small drone (Fig. 5(a)) or 11 m/s for the large drone (Fig. 5(b)), the RH and R2 models provide very similar short ranges. The more comprehensive R3 model provides a shorter range. Note that the calculated range values in Fig. 5 do not include any energy for hovering, takeoff and landing or ascending and descending, so the actual range would be less than that shown based purely on steady level flight (see Kirchstein, 2020; Xu, 2017).

Figs. 4 and 5 show that the *Epm* minimizing and range maximizing speeds are higher for large drones than small drones, and higher for R3 models than R2 models for the same drone. Table 7 shows the speed that minimizes *Epm* and the associated *Epm* and (maximum) range for the R2 and R3 models, along with results for the LD model at 12.5 m/s and the RH model at 6 m/s (these are the speed values used in the references presenting the LD and RH models). For R2 and R3, the large drone has a higher optimal speed and much greater *Epm* values (as expected). For the small drone at the optimal speeds, in comparison to the R2 model, the R3 model estimates a 34% higher *Epm* and a 25% shorter range. For the large drone at the optimal speeds, in comparison to the R2 model, the R3 model estimates a 96% higher *Epm* and a 49% shorter range. The LD model (with $r = 3$ at speed 12.5 m/s) generally provides a much lower *Epm* and a longer range than the other models (e.g., a 62% lower *Epm* and a 160% longer range for the small drone compared to R2). We note that the LD model uses $r = 3$ (from D'Andrea (2014)) which is not necessarily the best value for drone delivery with the small or large drone described in Table 6. These results underscore the importance of both model and parameter selection, especially the lift-to-drag ratio r for the LD model. For example, adjusting the lift-to-drag ratio r to a value of 0.8–1.2 (0.7–1.4) would make LD model produce results similar to the energy-minimizing values for the R2 and R3 models for small (large) drones.

4.2.2. The impact of payload on *Epm* and range for small and large drones

In this section, we examine the impact of payload on *Epm* and flight range for small and large drones described in Table 6 (the small quadcopter drone has a maximum payload of 0.5 kg and the large octocopter drone has a maximum payload of 7 kg). Because the *Epm* depends on payload and speed (See Table 5), we use the optimal (*Epm* minimizing) speed for the R2 and R3 models at each payload value as a fair comparison. For the R2 model with the small drone, the optimal speed is 9 m/s for payloads ranging from 0 to 0.47 kg, and 10 m/s for payloads ranging from 0.48 to 0.5 kg. For the R2 model with the large drone, the optimal speed is 13 m/s for payloads of 0–0.5 kg. For the R3 model, the optimal speeds are consistently 11 m/s and 16 m/s across payloads of 0–7 kg for small and large drones,

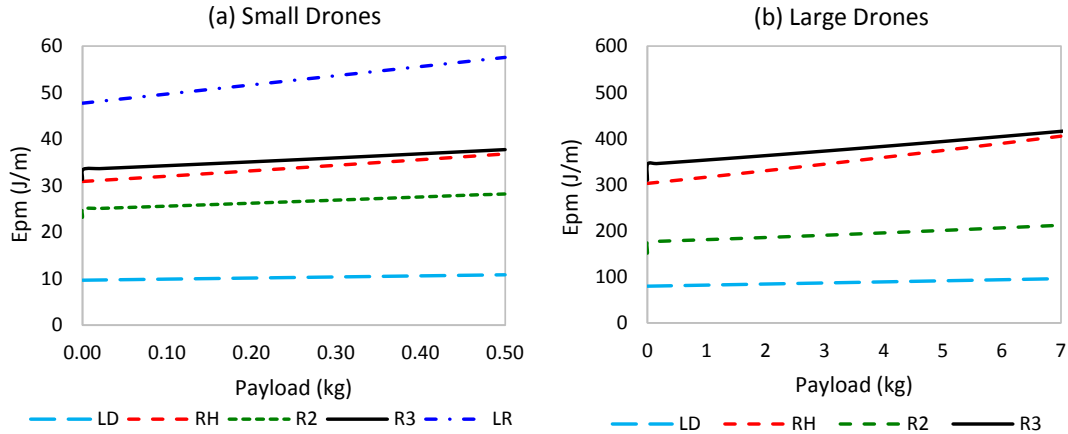


Fig. 6. Energy consumption rate versus payload for small and large drones.

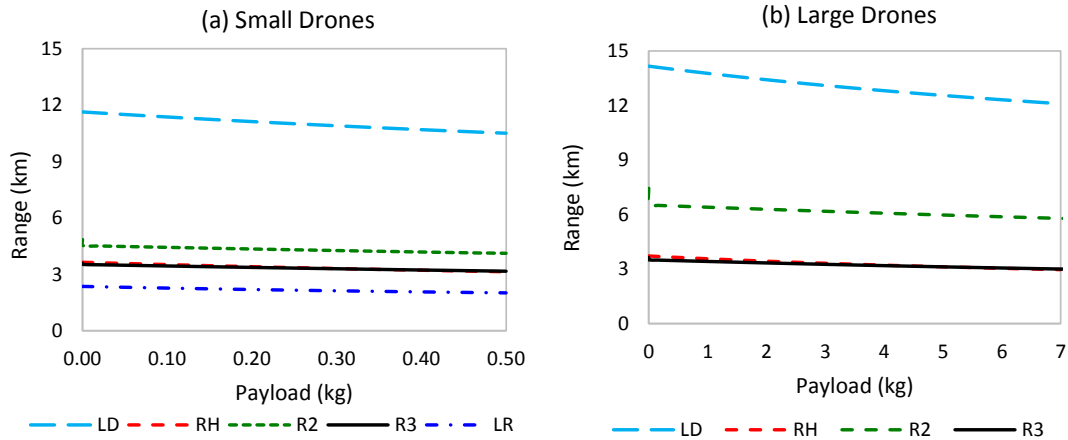


Fig. 7. Maximum range versus payload for small and large drones.

respectively. For the LD, RH and LR models, where E_{pm} is a point or a decreasing function of speed, we use the speed presented in the key reference (Table 5), i.e., 12.5 m/s for LD model, 6 m/s for RH model, and 5 m/s for LR model.

We begin by examining the effect of payload on E_{pm} . Fig. 6(a) and 6(b) show an approximately linear increase in E_{pm} with increasing payload for all models. In Fig. 6(b) the E_{pm} increases by 20–40% for the four models as the payload rises from 0 to 7 kg. While the patterns are similar in Fig. 6(a) and 6(b), the E_{pm} values for large drones are about 6–10 times greater than those of small drones for the same model within 0–0.5 kg payload range. This underscores the importance of using an appropriately sized drone for the payload. Using large drones with small payloads produces high E_{pm} values mainly due to the increased drone weight (including battery), which is outsized for the payload weight.

For both small and large drones, the LD model provides the lowest E_{pm} values (about 10 J/m and 80–96 J/m, respectively). For the small drone, the LR model provides the largest E_{pm} values of 47.8–57.6 J/m. The RH and R3 models provide similar values of 31.4–37.8 J/m and 344.0–415.3 J/m for small and large drones, respectively. The R2 model provides results which are approximately 2–2.5 times the LD results. As noted above, a smaller lift-to-drag ratio in the LD model would increase the E_{pm} to provide very similar results to the other models. We also observe that the high energy consumption for the R3 model is in part due to it including the profile drag, while using the same power transfer efficiency ($\eta = 0.7$) as the other models. So, modelers could use a lower power transfer efficiency to reflect modeling fewer components of power (e.g., a lower η value could be used in R2 compared to R3 for not including the profile power directly in R2). The high E_{pm} value of the LR for small drones (Fig. 6(a)) may reflect the importance of optimizing drone operating speeds as discussed in Section 4.2.1.

Fig. 7(a) and 7(b) show the flight range for all models as a function of the payload for small and large drones, respectively (again using optimal speeds for the R2 and R3 models). As expected, the flight range decreases for all models as payload increases. For both small and large drones, the LD model provides the largest ranges (10.5–11.6 km and 12.1–14.2 km, respectively) with the largest decreases (1.1 km or 11% and 2.1 km or 18%, respectively). As with E_{pm} , the LR model provides the smallest range values of 2.0–2.4 km for the small drone. The RH and R3 models provide similar values of 3.0–3.5 km for both small and large drones. The R2 model

provides results which are 39%–48% of the LD model.

As with E_{pm} , a smaller lift-to-drag ratio in the LD model, with r between 1.5 and 0.6 instead of 3, would produce range results between the lower limit from LR (or RH and R3) and the upper limit from R2. As with E_{pm} , the results show the large variability in ranges for the different models, a finding with clear implications for the number of customers that could be served by drone and for the number of drone depots needed to serve a region. In comparing the ranges of the small and large drones with the different models across the payload values, we observe the following: (i) for RH and R3, the range values are very similar for the large and small drone, (ii) for R2, the range for the large drone is 38–44% greater than that for the small drone. Note that the large drone has a battery with ten times more energy than the small drone, while the large drone total weight (body, battery and payload) is 8–9 times larger than the small drone depending on the payload.

In summary, Figs. 4–7 show the very wide variation in E_{pm} and range values for different modeling approaches with the same drone operating in a common setting. These differences have strong implications for accurately modeling the energy and assessing the environmental impacts of drone delivery. These results also indicate that the selection of the lift-to-drag ratio r and the power transfer efficiency η , both of which are difficult to assess without taking measurements in flight, can be crucial in accurately estimating drone energy consumption. It is also important to use an appropriate drone for the items (and purpose of delivery) and to optimize the operating speed.

5. Discussion and insights

The results in this paper demonstrate very large differences in drone energy consumption rates and operating ranges derived from different models in the literature. Some of the reported results differ due to different drone types and different operating conditions. However, even with a common setting that uses the same drone design and operating parameters (as in Section 4.2), the models provide E_{pm} and range estimates for steady level flight that differ substantially. Further, we note that when a full drone delivery flight profile is included (with takeoff, ascent, hovering, descent, and landing phases), the E_{pm} values can be substantially higher than the results for the steady level flight alone. Key findings from this research include:

- There are several fundamentally different approaches for modeling drone energy consumption, including integrated approaches reliant on a single lift-to-drag ratio to capture performance, component-based models developed to capture the influence of key aerodynamic forces, and regression models calibrated from field tests or from theoretical models.
- For published models, the energy consumption rate (E_{pm}) for steady level flight ranges from 16 J/m to over 400 J/m for different types of drones with different payloads. Even for small drones with payloads up to 1 kg, the literature provides E_{pm} values that differ by a factor of six. Note that many of these E_{pm} values reflect commercial off-the-shelf drones (for both theoretical modeling and field experiments), not drones designed specifically for package delivery, and thus may not reflect actual drone delivery operations.
- The large differences in E_{pm} values cannot be reconciled by using a common set of environmental, drone design and operating parameters. Even with the common sets of parameters, the energy consumption rate (E_{pm}) varies by a factor of 3–5 across the models (see Figs. 4 and 6). The estimated drone flight ranges differ similarly for the models.
- Integrated models (like the LD model) have the appeal of simplicity, but are very sensitive to the choice of the lift-to-drag ratio, which might vary with airspeed. Thus, only with a properly calibrated lift-to-drag ratio (based on field experiments in the relevant setting) and operating airspeed will they provide accurate estimates of energy consumption.
- Models based only on energy consumption for drone hovering (like the RH model) may provide very good approximations at low airspeeds. However, at higher speeds they cannot capture the increased parasite drag that grows to dominate the energy consumption.
- Component models provide the most detail, but are the most difficult to develop and calibrate given the number of parameters involved. Component models capture the strong dependence of E_{pm} and range on airspeed, and allow identification of an energy-minimizing airspeed.
- The two-component model (R2) and a three-component model (R3) highlighted the differences from modeling profile power directly and the importance of using a proper power transfer efficiency adjustment. All drone energy models include a power transfer adjustment to reflect losses due to a variety of sources, including battery charging efficiency, motor efficiency, drone blade performance, etc.; so it would be natural for a two component model to include a lower power transfer efficiency than a three component model to reflect power losses from the “missing” component. Our use of a common power transfer efficiency may explain in part the lower energy consumption for the R2 vs. R3 model.
- Published results for energy consumption from drone field tests often do not agree with results from theoretical energy models. We encourage modelers to account for the energy used in the entire drone delivery/trip profile (takeoff, ascent, hovering, descent, landing, and return), as well as for avionics and wind conditions to accurately estimate total energy consumption.
- There is a strong need for more field experiments to establish standards for drone energy consumption, analogous to vehicle fuel efficiency standards for trucks. This could include aggregate E_{pm} measures as well as parameter values for modeling various drone types, including new hybrid drones. Such a set of commonly accepted parameter values for effective delivery drones will be of great use in research on optimizing the design and operation of all types of drone operations, including delivery systems.

As energy consumption is important to both drone cost and emissions, an important broad research area is to better link drone energy consumption models with route optimization and strategic drone delivery systems design to assess the tradeoffs between cost,

Table A1
Manufacturer data for commercial drones.

Drone	Takeoff Weight (kg)	Max Flight Time	Max Hover Time (min)	Max Speed (m/s)	Max Flight Distance (km)	Distance Calculated from Column 3 (km)	Distance Calculated using Columns 4 & 5 (km)	Battery Energy (J)	Epm (J/m)
Mavic Spark	0.30	16 min at 5.56 m/s	15	13.9	9	5.33	12.5	60,739	6.7
Mavic Air	0.43	21 min at 6.94 m/s	20	19.0	10	8.75	22.8	98,753	9.9
Mavic Pro	0.73	27 min at 6.94 m/s	24	18.1	13	11.25	26.1	157,183	12.1
Mavic 2 Pro	0.91	31 min at 6.94 m/s	29	20.0	18	12.92	34.8	213,444	11.9
Phantom 4 Pro V2.0	1.38	30 min at 5.56 m/s	28	20.0	14	10.00	33.6	321,206	22.9
3DR-Solo	1.8	20 min	–	25.0	8	–	–	277,056	34.6
MD4-3000	11.0	45 min	–	20.25	36*	–	–	2,797,200	77.7

* from Figliozzi (2017).

energy, and emissions in drone delivery. For drones, the role of airspeed, as well as wind conditions, has a very important influence on energy consumption, more so than for trucks and other ground delivery vehicles, and should also be included in energy use models.

6. Conclusions

In this study, we provide an assessment of key energy consumption models for drone delivery. We identify the important factors that influence drone energy consumption and discuss and highlight key similarities and differences in drone energy models. We also provide an understanding of why the energy consumption models differ from each other, and identify important parameters that contribute to these differences. Our results document wide differences in drone energy consumption rates, even for models of the same drones applied in common settings. The selection of the lift-to-drag ratio r and the power transfer efficiency η , both of which are difficult to assess without taking measurements in flight, can be crucial in accurately estimating energy consumption for drones.

The energy consumption differences we document have strong implications for accurately modeling the energy and environmental implications of all drone operations, including delivery. Given that the models can provide Epm values that differ by a factor of 3–5 (or more), great care must be taken in translating results from transportation modeling (e.g., drone route modeling and optimization) to estimates and policy recommendations involving energy and emissions. The mixed research results regarding the energy and emissions efficiency of drone delivery are not surprising given the wide variability in energy consumption estimates produced from different Epm models. This underscores the need for further research to build a consensus on accurate parameter values for different types of delivery drones and different settings.

Our goal is not to identify one preferred model, but rather to document the discrepancies between models and highlight the need for accurate calibration of whatever model is adopted. All models examined in this paper can produce similar results with an appropriate set of parameter values. Given the lack of widespread drone delivery operations that limit direct measurements in the field, it is not surprising there is no agreement on accurate parameter values in the academic research community. Further, with the rapid rate of evolution in drone technology, those parameter values will likely change frequently as the field matures, necessitating ongoing research.

This research suggests a number of important areas for future research. Clearly, a better understanding of the accuracy of drone energy models is needed through comparing results to empirical data derived from comprehensive drone delivery field tests. These empirical tests might best be undertaken in a partnership between government agencies (e.g., the U.S. Department of Energy or EPA), academic institutions, and private sector firms. The importance of avionics and wind conditions on drone energy consumption is an area especially needing more attention, with particular care to the numbers, sizes and types of drones deployed. Future research can also help identify which type of model (complex or simple) is best in different settings, and whether or when more parsimonious models are “accurate enough” to use.

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Appendix A

With appropriate data, the drone energy consumption rate Epm can be calculated as

$$E_{pm} = \frac{BatteryEnergy}{FlightDistance} \quad (A1)$$

Information on the *BatteryEnergy* is often, but not always, provided by the manufacturer; either in Watt-hours (1 Wh = 3600 J) or as capacity and voltage, where

$$BatteryEnergy(inJ) = Capacity(inmAh) \times Voltage(inV) \times 3.6. \quad (A2)$$

However, *FlightDistance* information is not reported by all manufacturers, nor is it provided in a consistent manner. Flight distance data from manufacturer websites may be the distance under conditions that maximize that distance (e.g., the speed, altitude and weather conditions that maximize distance) and do not reflect real-world drone operations. We are aware of only a few drones for which flight distance at an associated speed is reported (see Table A1). As most manufacturers report a flight time or hovering time and a maximum speed for each drone, it is tempting to use that data to estimate flight distance as

$$FlightDistance = FlightSpeed \times FlightTime \quad (A3)$$

Yet, because manufacturers often report a maximum value (e.g., “Max Flight Time” or “Max Speed”) or an unspecified value (e.g., “Flight Time”), and drone energy use depends strongly on speed, combining a maximum flight (or hovering) time with a maximum speed is likely to provide an inaccurate flight distance measure. Thus, the seemingly straightforward approach of *FlightSpeed* × *FlightTime* is fraught with uncertainties due to unclear and inconsistent reporting of information by different manufacturers, as well as inconsistencies across drones for the same manufacturer.

As an example of the difficulties, consider the data below for DJI drones in the first 5 rows of Table A1 (from <https://www.dji.com/products/compare-consumer-drones>), which provides “Max Flight Time” at a particular speed (see column 3), along with “Max Hovering Time”, “Max Speed” and “Max Flight Distance” (see columns 4–6). The flight distance calculated based on the time and speed in column 3 using eq. (A3) is shown in column 7, and the flight distance calculated based on the hovering time (column 5) and maximum speed (column 6) (using eq. (A3) with hovering time) is shown in column 8. Note that the three flight distance values in columns 6–8 disagree considerably. It is not surprising that column 8 is much greater than the flight distance in column 6 (on average about twice as large), because high speeds increase the rate of energy consumption which would reduce the endurance at that maximum speed to well below the hovering time. It may also not be surprising that the distances calculated in column 7 (based on the speed and time from column 3) are much less (13%–40% less) than the reported flight distance in column 6, if the column 6 values are based on the distance-maximizing speed (while column 3 may report the typical operational speed; and we note speeds of 5–6 m/s are typical for imaging applications). This illustrates the need to use common data elements, and the dangers of combining different calculations of flight distance across different manufacturers.

We do not include values for other DJI drones (Inspire, the Matrice series, and the Agricultural drones AGRAS T-16 and the MG series) in Table A1, as the website does not provide flight distances for its other drones (only the maximum flight or hovering time and maximum speed). We were able to collect relevant data for 2 other drones from manufacturer websites, as shown in the last two rows of Table A1 for the Microdrones MD4-3000 and the 3D Robotics 3DR-Solo. Thus, Table A1 includes data for three manufacturers where both flight distance and battery energy information was reported. Columns 2–6 and 9 are from the manufacturer websites. Column 9 is the battery energy, either provided directly or calculated based on capacity (mAh) × voltage. Columns 7 and 8 are calculated flight distances for the DJI drones as described above. Column 10 is the *Epm* calculated using equation (A1) (column 9 divided by column 6).

Because of the issues illustrated above for the DJI drones, we are hesitant to recommend computing energy consumption rates using manufacturer data unless the drone operating conditions are clearly specified and they match the particular application of interest (e.g., imaging may require slower speeds than delivery). We also note that energy consumption rates calculated with manufacturer data using a takeoff weight with no payload is not directly comparable to energy consumption rates in the literature for delivery drones carrying a payload. Finally, we also note that some researchers reduce the battery energy available for flight or the travel distance to reflect operating realities. For example, Figliozzi (2017) assumes for one of his analyses that the drone flight range is only 70% of the maximum flight distance to account for unknown factors (e.g., bad weather) that increase energy consumption and to provide a safety margin. To achieve a similar goal, Stolaroff et al. (2018) and Troudi et al. (2018) assume a safety factor of 1.2 and 1.25, respectively, to reserve energy for unusual conditions (which is equivalent to using 83.3% and 80% of maximum flight distance, respectively).

Appendix B

This appendix provides a derivation of the energy consumption for a battery powered aircraft in steady level flight (i.e. at a constant altitude and constant speed), as in D’Andrea (2014), based on fundamental principles of flight. Four fundamental forces acting on an aircraft in steady flight are thrust to move forward, weight from gravity acting on the aircraft mass, lift in the direction opposing gravity, and drag in the direction opposing travel (from aerodynamic effects of the shape of the aircraft, such as friction with air). The amount of energy in joules needed to fly a distance d (measured in meters) at constant altitude can be modeled as

$$E_{fly} = T \times d \quad (B1)$$

where T is the thrust force measured in Newtons (kg-m/sec²). Note that this does not account for the energy involved with the vertical travel components to lift the aircraft from the ground to a cruising altitude and to lower it back to the ground. For drone delivery, this may occur twice each trip, once at takeoff/landing and once for the delivery; however, drone deliveries may not require landing, as the package may be lowered via a tether or dropped via parachute.

An important factor for aircraft performance is the lift-to-drag ratio r , a unitless parameter which captures the efficiency of the aircraft design in keeping the aircraft airborne. Values of the lift-to-drag ratio range widely from about 10–20 for commercial passenger aircraft to about 4 for helicopters in cruising flight, to typically smaller values for UAV rotocopters. The lift-to-drag ratio varies with aircraft speed due to drag and the aerodynamic affects from lifting surfaces, but is constant in steady flight. To keep the aircraft in steady flight, lift is equal to weight and thrust is equal to drag, so

$$T = \text{drag} = \frac{\text{lift}}{r} = \frac{\text{weight}}{r} = \frac{\text{mass} \times g}{r} \quad (\text{B2})$$

where g is the acceleration due to gravity (9.8 m/sec²) and mass is the total weight of the aircraft (body, battery and payload) in kg. Combining (B1) and (B2) gives

$$E_{\text{fly}} = \frac{(m_1 + m_2 + m_3) \times g}{r} \times d \quad (\text{B3})$$

where m_1 is the mass of the aircraft structure, m_2 is the mass of the aircraft battery and m_3 is the mass of the payload.

An important parameter for battery powered drones is the efficiency for converting battery power to “flight” power delivered by the rotors, denoted η (unitless), where $\eta < 1$. Thus, the energy required from the battery for steady flight is

$$E_{\text{fly}} = \frac{(m_1 + m_2 + m_3) \times g}{\eta r} \times d \quad (\text{B4})$$

Additional energy is consumed by the drone avionics required for safe flight, including communications, sensing, computation, etc. Let P_{avio} be the power required for the aircraft (drone) avionics in joules per second of flight. The energy for avionics (in joules) for a flight of distance d is then

$$E_{\text{avio}} = P_{\text{avio}} \times \frac{d}{v} \quad (\text{B5})$$

The total energy expended by the aircraft battery (in joules) on a flight of distance d is from equations (B4) and (B5):

$$E_{\text{tot}} = \frac{(m_1 + m_2 + m_3)g}{\eta r} d + P_{\text{avio}} \times \frac{d}{v} \quad (\text{B6})$$

The total energy expended per meter of flight E_{pm} is then

$$E_{\text{pm}} = \frac{(m_1 + m_2 + m_3)g}{\eta r} + \frac{P_{\text{avio}}}{v} \quad (\text{B7})$$

This is equivalent to the formula presented in D’Andrea (2014) for power

$$\frac{(m_1 + m_2 + m_3)v}{370r\eta} + P_{\text{avio}}$$

because power equals $E_{\text{pm}} \times v$, and the constant 370 in the denominator of the first term results from substituting in 9.81 for g and measuring speed in km/hr rather than m/sec (3600/9.81 = 367).

Appendix C

The model in Kirchstein (2020) for steady flight is based on the power model in Langelaan et al. (2017) where the power for constant speed flight includes four components for: induced power for lift, power to overcome parasite drag, profile power, and ascending/descending power (based on flight angle θ). This is expressed as a function of the mass and flight angle:

$$P(m_k, v_a, \gamma) = \kappa w T + \frac{1}{2} \rho \left(\sum_{k=1}^3 C_{D_k} A_k \right) v_a^3 + \frac{\rho \sqrt{\zeta} / \pi n N c c_d}{8} v_T^3 \left(1 + 3 \left(\frac{v_a}{v_T} \right)^2 \right) + \left(g \sum_{k=1}^3 m_k \right) v_a \sin \theta \quad (\text{C1})$$

where $v_T = \sqrt{\frac{6}{n N c c_d \rho \sqrt{\zeta} / \pi}} \sqrt{g \sum_{k=1}^3 m_k}$ is the blade tip speed. The parameter w , like the induced velocity in Stolaroff et al. (2018), is found by solving the following equation

$$w = \frac{T}{2n\rho\zeta\sqrt{(v_a \cos \alpha)^2 + (v_a \sin \alpha + w)^2}} \quad (\text{C2})$$

where T is from equation (20) and α is from equation (15).

The energy per meter for steady level flight ($\theta = 0$) for a total drone mass of m (including battery and payload if any) with avionics power P_{avio} is

$$E_{\text{pm}} = \frac{P(m, v_a, 0)}{\eta v_a} + \frac{P_{\text{avio}}}{\eta_c v_a} \quad (\text{C3})$$

Kirchstein (2020) uses a total energy efficiency η for drone flight power that includes the motor, transmission and battery charging efficiency, but only the charging efficiency η_c for the avionics.

For a delivery by a drone moving at speed v_a with no wind to a distance d from the depot, the total energy including take-off and ascent at 45° to an altitude of alt , level flight, hovering for time t_{hover} , and then landing, is

$$E_{pm} = \frac{1}{v_a} \left[\frac{P(m, v_a, 0)}{\eta} + \frac{P_{avio}}{\eta_c} \right] + \frac{1}{v_a} \frac{alt}{d} \frac{1}{\eta} [P(m, v_a, 45^\circ) + P(m, v_a, -45^\circ) - 2P(m, v_a, 0)] + \frac{t_{hover}}{d} \left[\frac{P(m, 0, 0)}{\eta} + \frac{P_{avio}}{\eta_c} \right] \quad (C4)$$

The return trip would be the same, but without the payload ($m = m_1 + m_2$). The first term in (C4) is for the level flight, which will usually be the majority of the trip (unless the delivery is for a very short distance), the second term is for the ascending and descending, and the last term is for hovering.

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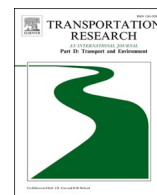
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Update

Transportation Research Part D

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Corrigendum



Corrigendum to “Energy consumption models for delivery drones: A comparison and assessment” [Transp. Res. Part D: Transp. Environ. 90 (2021) 102668]

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The authors regret that the following corrections need to be made for this article, and apologize for any inconvenience caused. However, it does not affect the results of the analyses or the conclusions from the original contribution.

The authors mistakenly interpreted the drone used in Kirschstein (2020) to have a body weight of 2 kg. The correct value is a body weight of 12 kg. The following are the corrections for the weight and the associated energy consumption reported for the Kirschstein (2020) drone.

In Section 3.3, the following sentence needs to be corrected: ‘Kirschstein (2020) provides analyses for delivery in an urban region (comparing drones with diesel and electric trucks) with large octocopter drones ($m_1 + m_2 = 12$ kg) that carry a 2.5 kg payload ($m_3 = 2.5$ kg), travel at 22.2 m/s, have a flight radius of 9 km (and the trips include 5 min of hovering), power transfer efficiency $\eta = 0.73$, battery charging efficiency $\eta_c = 0.9$ and $P_{avio} = 100$ J/s (as in D’Andrea (2014)). For this drone, the constants in eq.(20) are: $\kappa = 1.15$, $\kappa_2 = 0.502$ and $\kappa_3 = 0.118$; and the loaded $E_{pm} = 131.5$ J/m’. The value for the large octocopter drones should be $m_1 + m_2 = 22$ kg. The value for loaded E_{pm} should be $E_{pm} = 225.8$ J/m.

In Table 3a, the entry in column 6 (heading “ $m_1 + m_2$ (kg)”) for Kirschstein (2020) should be **22** instead of 12.

In Table 3b, the entry in column 3 (heading “ E_{pm} (J/m)”) for Kirschstein (2020) should be **213.9** instead of 127.2, and the entry in column 5 (heading “Body Weight (kg)”) should be **12** instead of 2.

In Fig. 3, the energy consumption per meter of point K-8-L should be **208.9** instead of 117.6, so the correct figure is shown below.

In Section 4.1, the following sentence should be corrected: ‘The next four rows for Kirschstein (2020) show a very small increase (3.4%) in E_{pm} from including avionics, a very large increase (129%) due to including strong winds of 45 km/hr, and a substantial increase (70.1%) by including both moderate winds (10 km/hr) and energy use for a complete flight profile’. The correct sentence is: ‘The next four rows for Kirschstein (2020) show a very small increase (**2.4%**) in E_{pm} from including avionics, a very large increase (**134%**) due to including strong winds of 45 km/hr, and a **moderate** increase (**22.4%**) by including both moderate winds (10 km/hr) and energy use for a complete flight profile’.

In Table 4, the entry in column 4 (heading “ $m_1 + m_2$ (kg)”) for Kirschstein (2020) should be **22** instead of 12; the entry in column 8 (heading “ E_{pm} (J/m)”) should be **208.9** instead of 127.2 for Baseline, **213.9** instead of 131.5 for Avionics, **489.2** instead of 300.6 for Wind (45 km/h) and avionics, and **255.7** instead of 216.4 for a complete flight profile with wind (10 km/h) and avionics; the entry in column 9 (heading “% E_{pm} increase to baseline”) should be **2.4%**, **134%**, and **22.4%**, instead of 3.4%, 129%, and 70.1%, respectively. (Since Table 4 has more than one number to correct, we show the corrected entries for Kirschstein (2020) as below.

The authors would like to apologize for any inconvenience caused.

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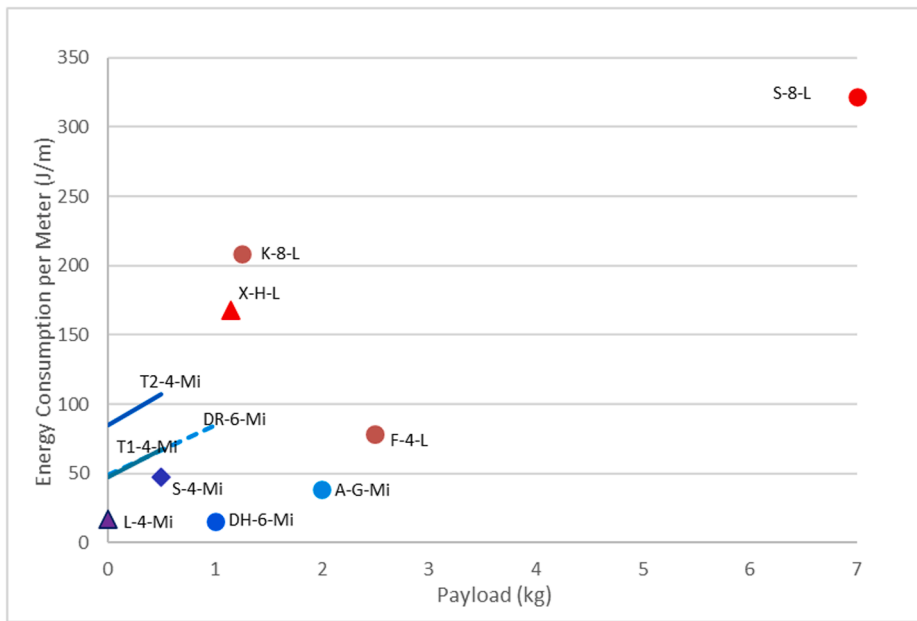


Fig. 3. *Epm* Results from baseline models in Tables 3a and 3b.

Table 4
Key drone energy models with additional conditions.

ID	Reference	Additional conditions to baseline	$m_1 + m_2$ (kg)	Payload ¹ (kg)	Airspeed v_a (m/s)	Range ² (km)	E_{pm} (J/m)	% E_{pm} increase to baseline
K-8-L	Kirschstein (2020)	Baseline	22	1.25 ¹	22.2	9	208.9	–
		Avionics					213.9	2.4%
		Wind (45 km/h), avionics					489.2	134%
		A complete flight profile with wind (10 km/h) & avionics					255.7	22.4%